

*MA.6.GR.2.4***Benchmark**

MA.6.GR.2.4 **Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.**

Benchmark Clarifications:

Clarification 1: Instruction focuses on representing a right rectangular prism and right rectangular pyramid with its net and on the connection between the surface area of a figure and its net.

Clarification 2: Within this benchmark, the expectation is to find the surface area when given a net or when given a three-dimensional figure.

Clarification 3: Problems involving right rectangular pyramids are limited to cases where the heights of triangles are given.

Clarification 4: Dimensions are limited to positive rational numbers.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.2, MA.6.NSO.2.3
- MA.6.AR.2.3

Terms from the K-12 Glossary

- Area
- Net
- Rectangular Prism
- Rectangular Pyramid

Vertical Alignment**Previous Benchmarks**

- MA.5.GR.2.1

Next Benchmarks

- MA.7.GR.2.2

Purpose and Instructional Strategies

In previous courses, students found the area of a rectangle with fractional or decimal sides. In grade 6 accelerated, students find the surface area of right rectangular prisms and right rectangular pyramids. In future courses, students will calculate the surface area of right circular cylinders.

- Instruction includes constructing models and nets of three-dimensional figures and describing them by the number of edges, vertices, and faces. Providing these opportunities will allow students to manipulate materials and connect to more abstract aspects of geometry.
- Instruction includes students using formulas and decomposition strategies to find the surface area of figures. Using the dimensions of the individual faces, students should calculate each rectangle or triangle area and add these values together to find the surface area of the figure.
- Instruction includes describing the types of faces needed to create a three-dimensional figure. Students can make and test conjectures by determining what is needed to create a specific three-dimensional figure.

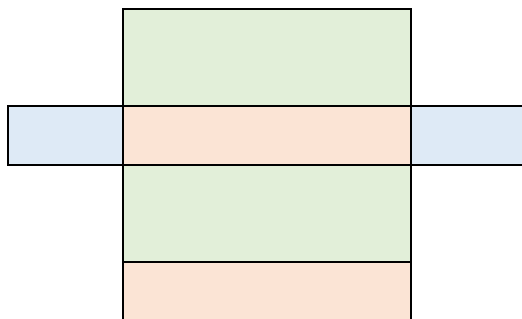
- Students need to practice the language of the problems they are being asked to solve.
 - For instance, when being asked how much wrapping paper is needed to wrap a box, they know from experience that they are working with surface area. If they are being asked how many boxes will fit in a shipping container, then they are looking at volume.
- When using rational numbers, instruction should stay within the same form. Students should not be penalized though if they convert from one form to another when performing operations.
 - For example, if students are working with fractions, the side lengths will not include decimals. If students are working with decimals, the side lengths will not include fractions.
- Instruction includes representing measurements for surface area as square units, units squared or units^2 .
- Problem types include having students measure lengths using a ruler to determine the surface area.

Common Misconceptions or Errors

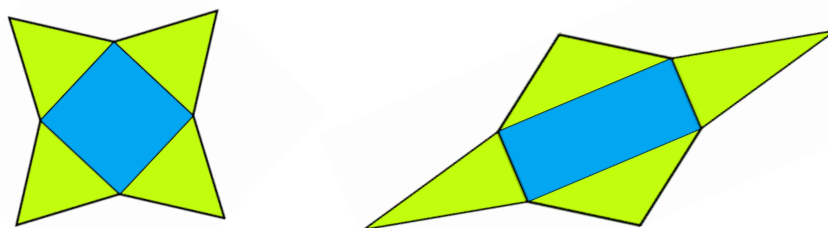
- Students may not be able to determine the different two-dimensional figures that make up the net of the three-dimensional figure.
- Students may invert the formulas for surface area and volume.
- Students may confuse rectangular prisms and right rectangular pyramids.

Strategies to Support Tiered Instruction

- Teacher provides nets of the three-dimensional figures and model color coding each similar shape. This will help students properly identify each shape to find the area to calculate surface area.
 - For example, a right rectangular prism can be modeled by the net shown below.



- For example, a right rectangular pyramid can be modeled by the net shown below.

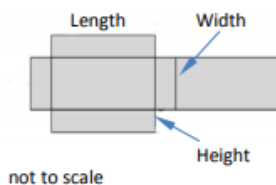


- Teacher reviews definitions of surface area and volume and co-creates an anchor chart to display in the room explaining the differences between them. Teacher models the use of manipulatives and geometric software to review the concept of area and surface area.
- Teacher breaks down formulas for area of a rectangle and volume of a rectangular prism to show when finding area, we are multiplying two sides which is why we use units², but with the rectangular prism, we are multiplying three sides, so we use units³ to label. Providing flashcards or cue cards with the formulas will help students in place of anchor charts when they are outside the classroom area.

Instructional Tasks

Instructional Task 1 (MTR.6.1)

The surface area of a rectangular prism is 115 square inches. The net of the prism is shown. What are the possible dimensions of the prism?



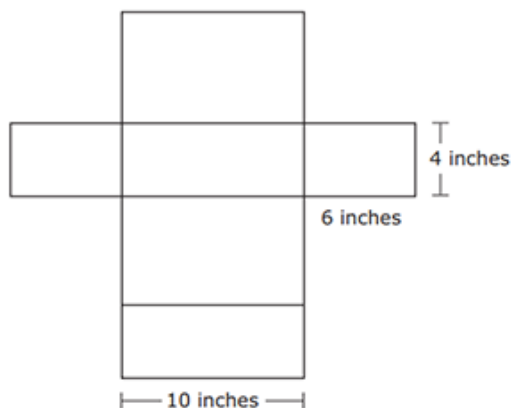
Instructional Task 2 (MTR.2.1)

A right rectangular pyramid has a surface area of 56cm^2 and the height of each triangular side is 5cm. Draw the net of the figure described.

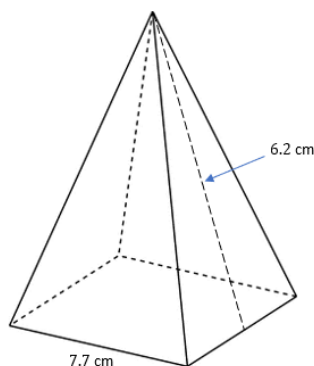
Extension. If the base of the pyramid is a square, what are the dimensions of the base?

Instructional Items*Instructional Item 1*

Carl is making a box to ship a gift. The net of Carl's box is shown. How much cardboard, in square inches, is required to make the box?

*Instructional Item 2*

Maxwell is making a replica of the Egyptian pyramids. The figures' dimensions have a square base with a side length of 7.7 cm and a slant height of 6.2 cm. Maxwell wants to cover the model with gold paper. How much paper, in square centimeters, is needed to cover the pyramid?



**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*